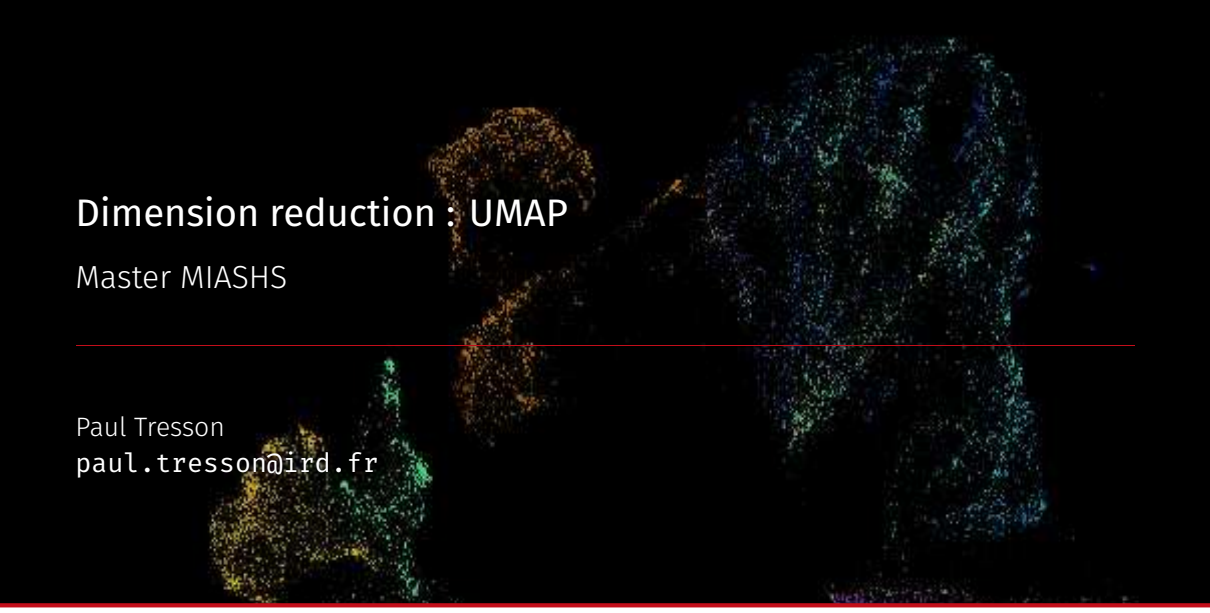


Dimension reduction : UMAP

Master MIASHS

Paul Tresson
paul.tresson@ird.fr

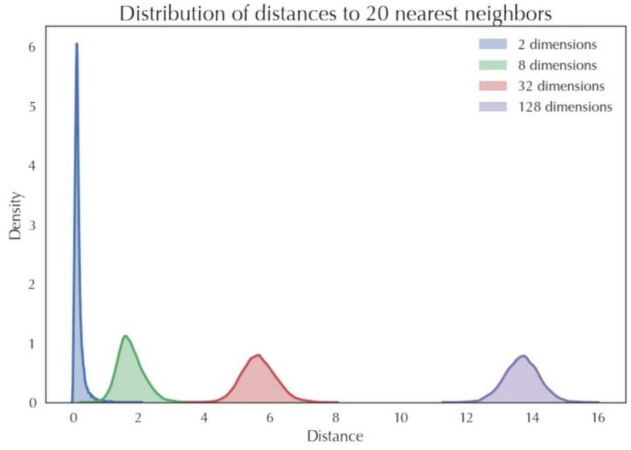
Institut de Recherche pour le Développement FRANCE
AMAPlab
UNIVERSITÉ PAUL VALÉRY MONTPELLIER

Why reduce dimensions ?

Why reduce dimensions ?

- Visualization and exploration
- Dimensions can be redundant
- Dimensions can be non-informative
- **Curse of dimensionality**

Curse of dimensionality



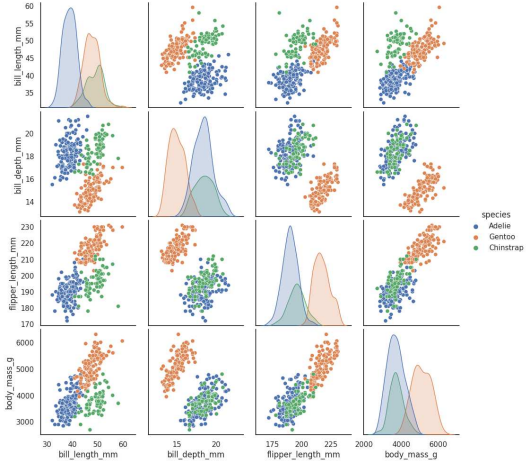
Leland McInnes slides

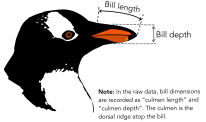
Working with less dimensions can help

- Clustering
- Outlier detection
- Scaling up
- ...

Visualize what is happening in higher dimensions



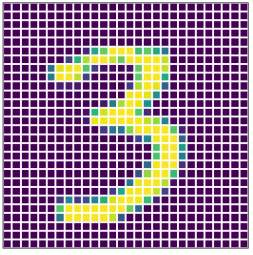




Visualize what is happening in higher dimensions

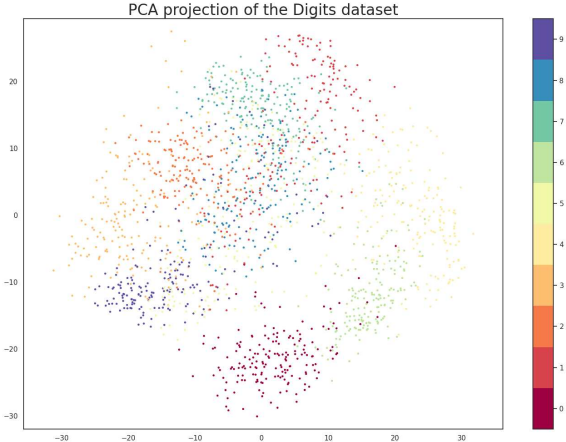


Visualize what is happening in higher dimensions

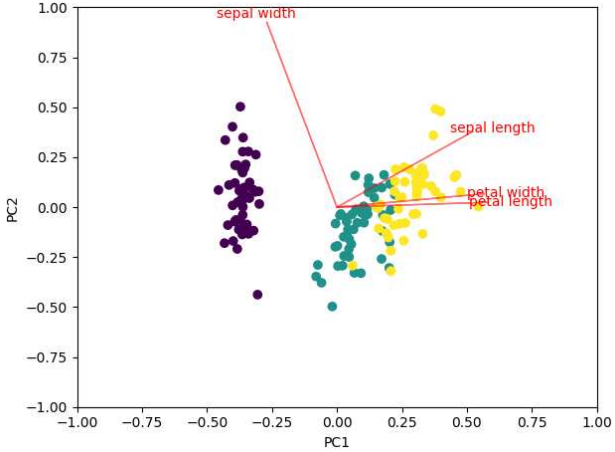


label	pixel 001	...	pixel 350	pixel 351	pixel 352	...	pixel 728
3	0		253	253	187		0
1	0		228	245	51		0
⋮	⋮		⋮	⋮	⋮		⋮

PCA


PCA projection of the Digits dataset
20
10
0
−10
−20
−30
−30
−20
−10
0
10
20
30
9
8
7
6
5
4
3
2
1
0

PCA interpretability



PCA advantages and drawbacks

Advantages

- fast

Drawbacks

PCA advantages and drawbacks

Advantages

- fast
- scales well

Drawbacks

PCA advantages and drawbacks

Advantages

- fast
- scales well
- explainatory

Drawbacks

PCA advantages and drawbacks

Advantages

- fast
- scales well
- explainatory

Drawbacks

- not suited for non-linear data

Non linear data

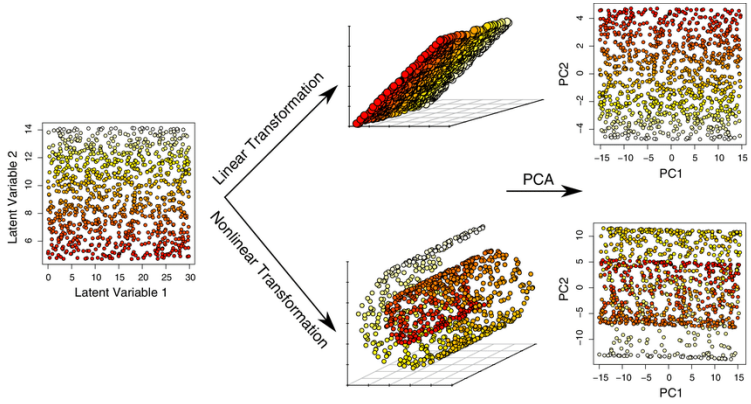
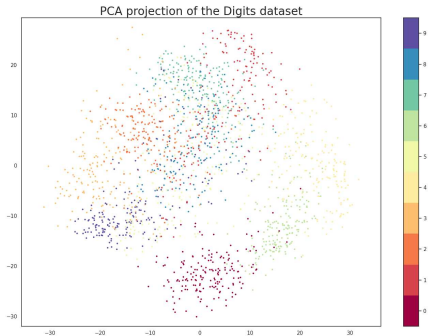


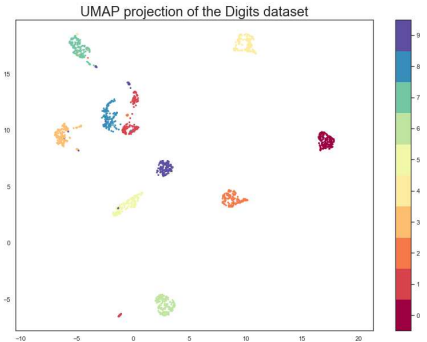
Figure from Du, 2019

What is a manifold ?

In mathematics, a **manifold** is a topological space that locally resembles Euclidean space near each point. More precisely, an n-dimensional manifold, or *n-manifold* for short, is a topological space with the property that each point has a neighborhood that is homeomorphic to an open subset of n-dimensional Euclidean space.

Non linear data





How to work with non linear data

1. Find a good representation of the data in high dimension
2. Fit a good representation of in low dimension

Classics

- **SNE** (Stochastic Neighbor Embedding) Hinton and Roweis, 2002

Classics

- **SNE** (Stochastic Neighbor Embedding) Hinton and Roweis, 2002
- **T-SNE** (T-distributed SNE) Van der Maaten and Hinton, 2008

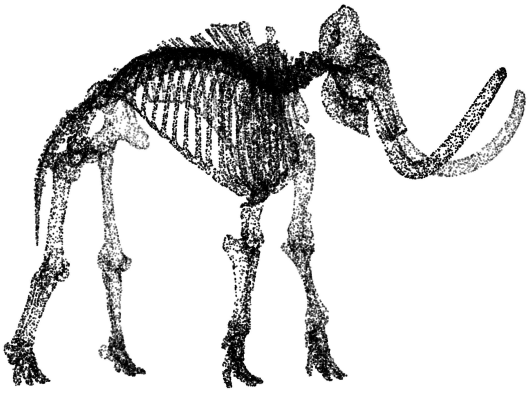
Classics

- **SNE** (Stochastic Neighbor Embedding) Hinton and Roweis, 2002
- **T-SNE** (T-distributed SNE) Van der Maaten and Hinton, 2008

- **UMAP** (Uniform Manifold Approximation and Projection) McInnes et al., 2018

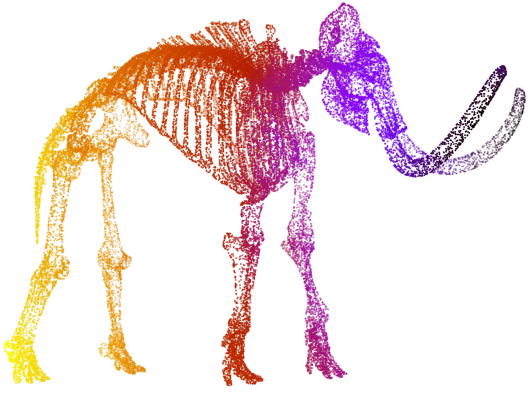
How does UMAP work ?

Mammoth example



Mammoth dataset by the Smithonian

Mammoth example

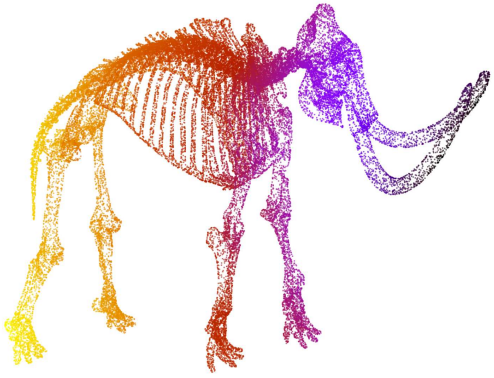


Mammoth dataset by the Smithonian

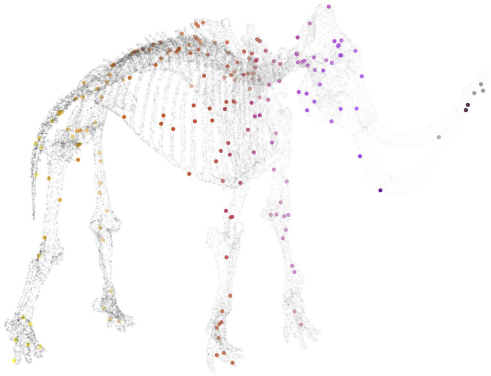
PCA



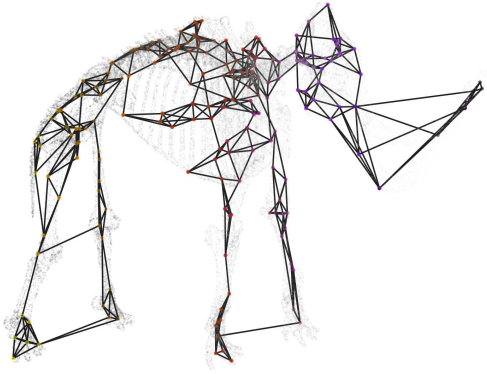
Finding high dimension graph



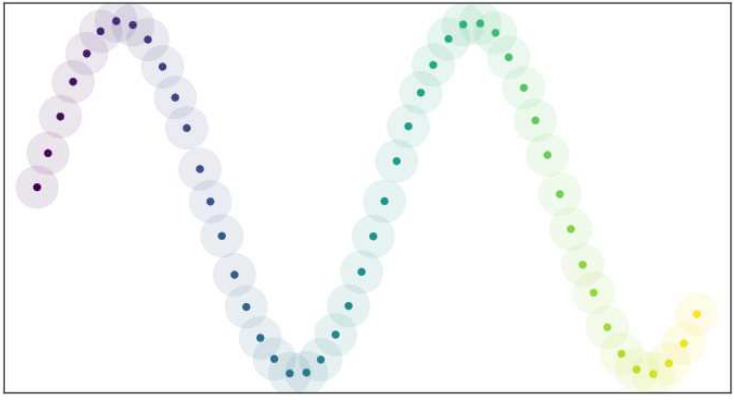
Finding high dimension graph



Finding high dimension graph

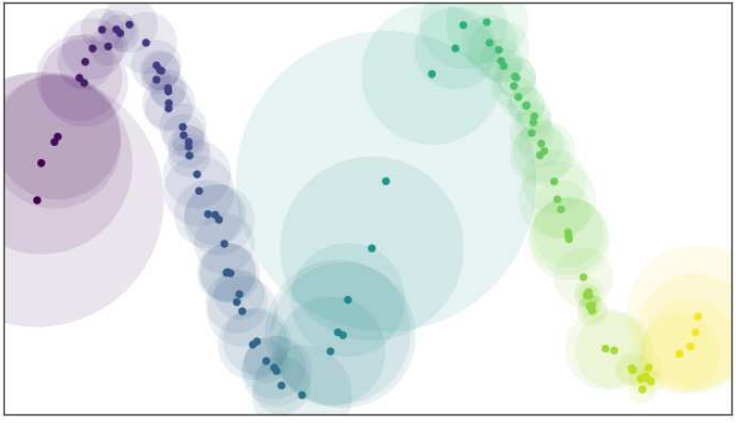


Uniform Manifold Approximation and Projection



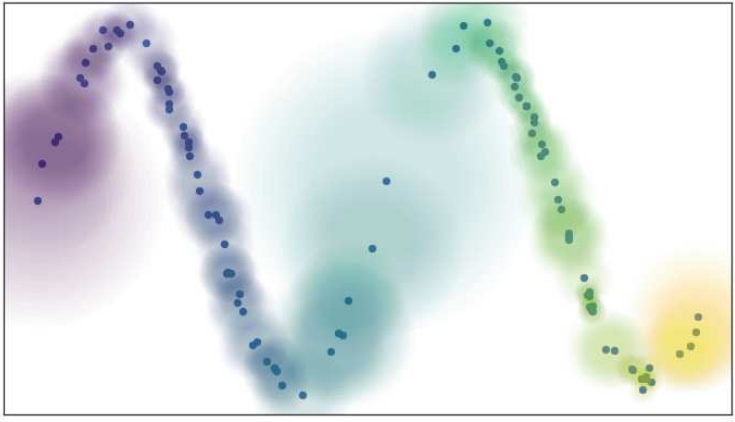
UMAP documentation

Uniform Manifold Approximation and Projection



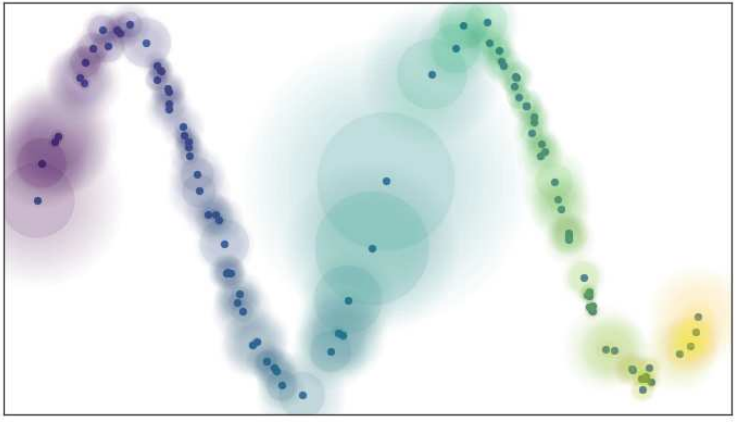
UMAP documentation

Uniform Manifold Approximation and Projection



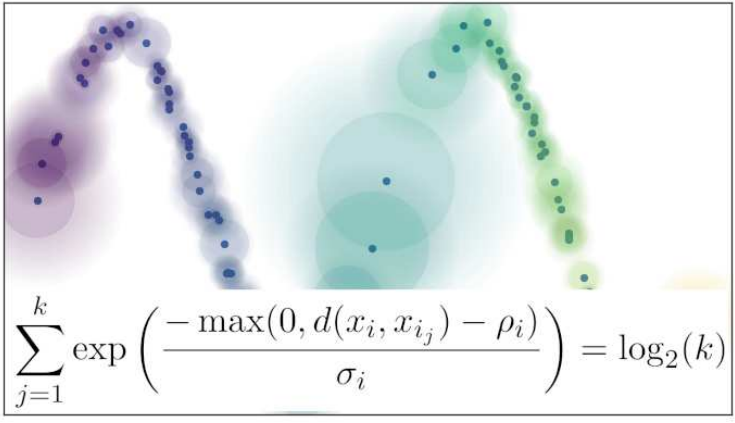
UMAP documentation

Uniform Manifold Approximation and Projection



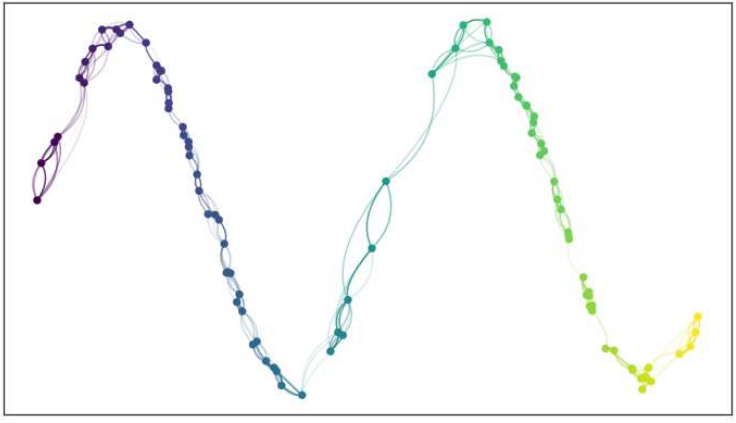
UMAP documentation

Uniform Manifold Approximation and Projection



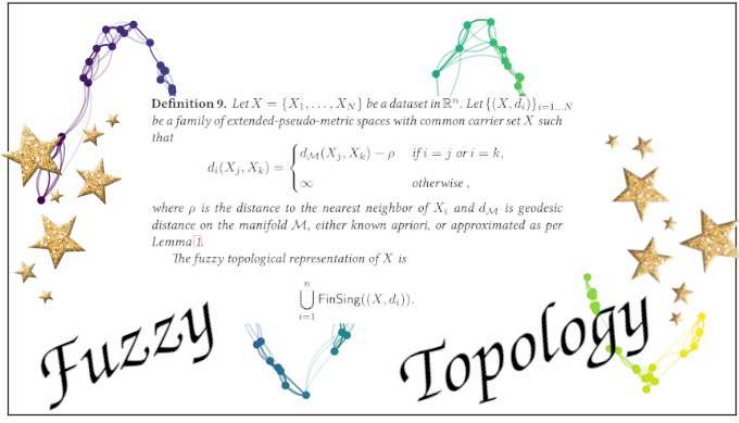
UMAP documentation

Uniform Manifold Approximation and Projection



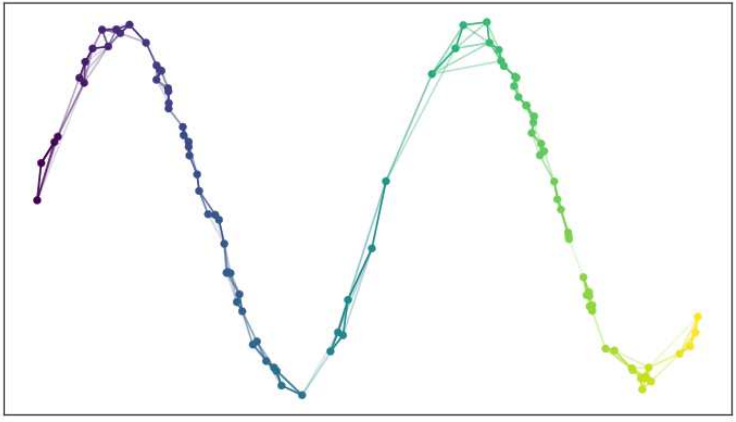
UMAP documentation

Uniform Manifold Approximation and Projection



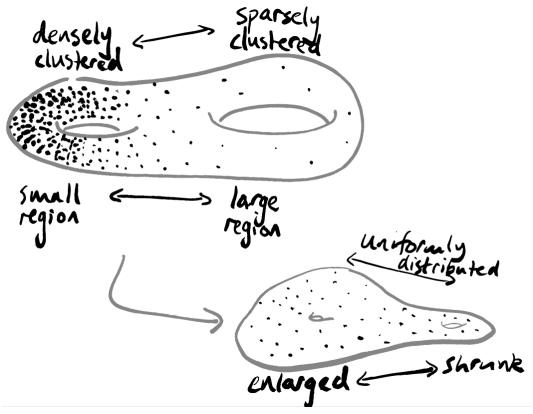
UMAP documentation

Uniform Manifold Approximation and Projection



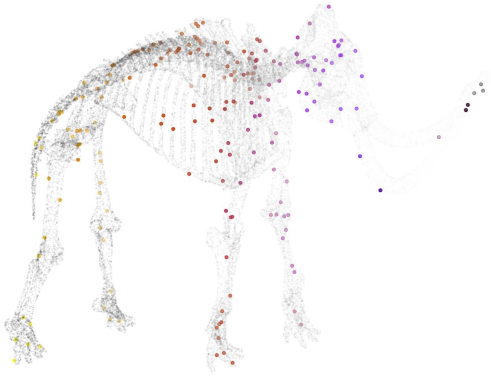
UMAP documentation

Uniform Manifold Approximation and Projection

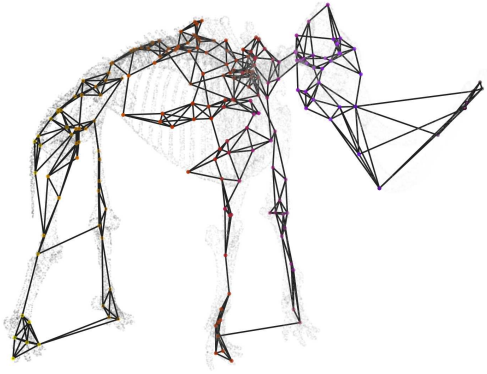


Tim Hosgood - Understanding UMAP

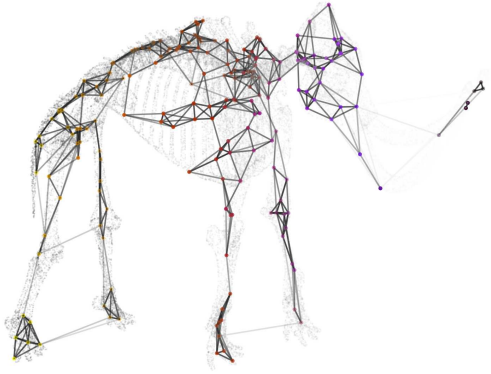
Finding high dimension graph



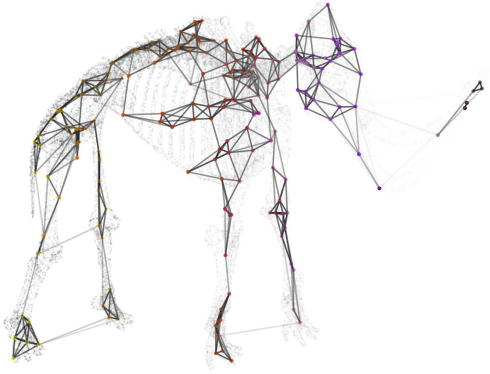
Finding high dimension graph



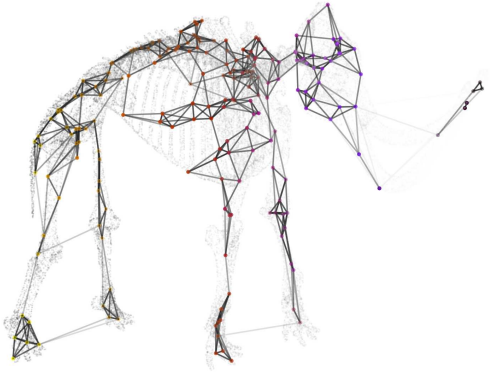
Finding high dimension graph

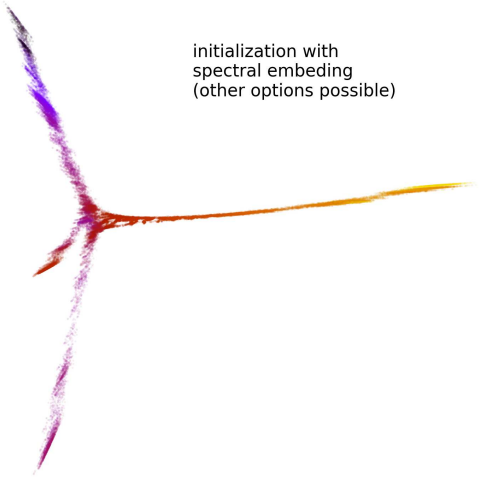


Fitting low dimensional representation



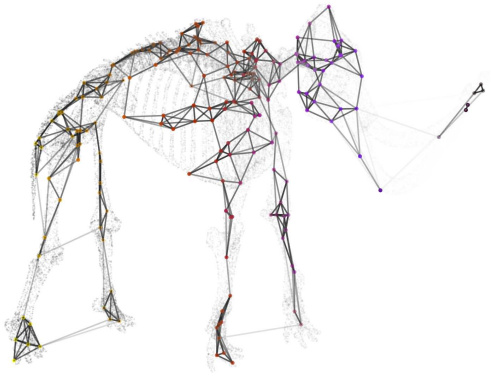
Fitting low dimensional representation

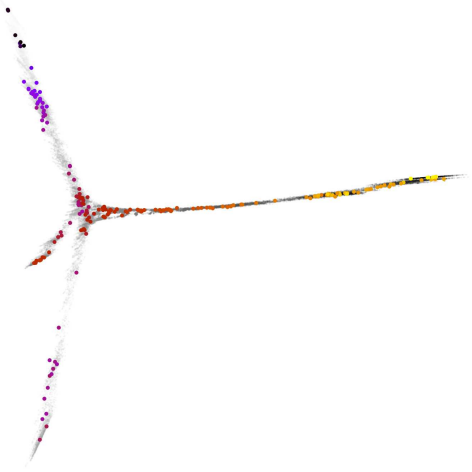




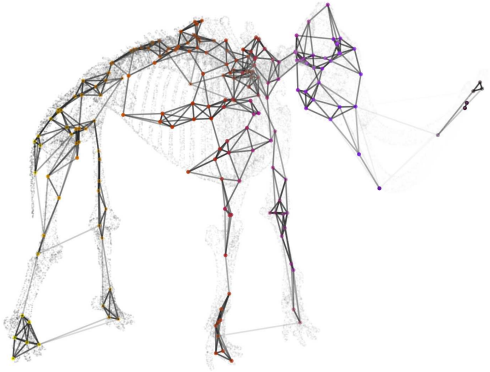
initialization with
spectral embeding
(other options possible)

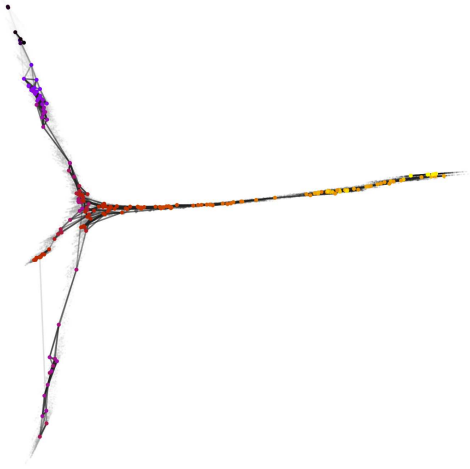
Fitting low dimensional representation



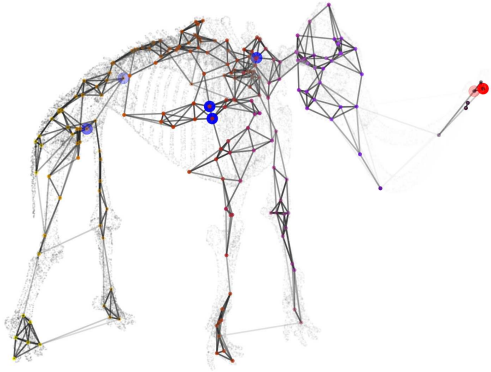


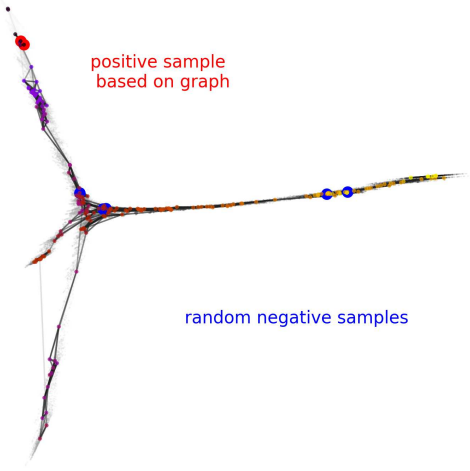
Fitting low dimensional representation





Fitting low dimensional representation





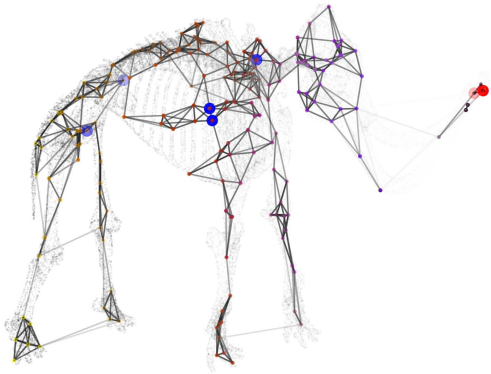
Fitting low dimensional representation

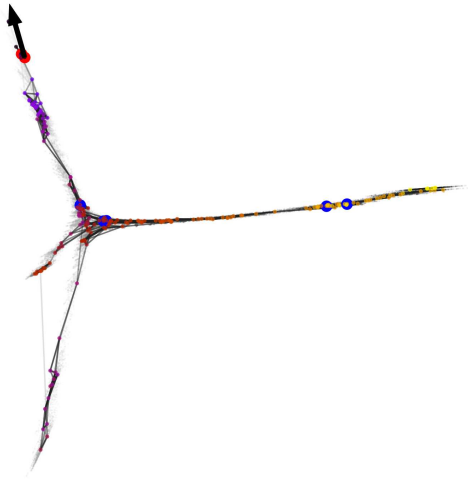
Cross-entropy loss counting positive and negative atraction forces

$$C_{UMAP} = \sum_{i \neq j} v_{ij} \log\left(\frac{v_{ij}}{w_{ij}}\right) + (1 - v_{ij}) \log\left(\frac{1 - v_{ij}}{1 - w_{ij}}\right) \quad (1)$$

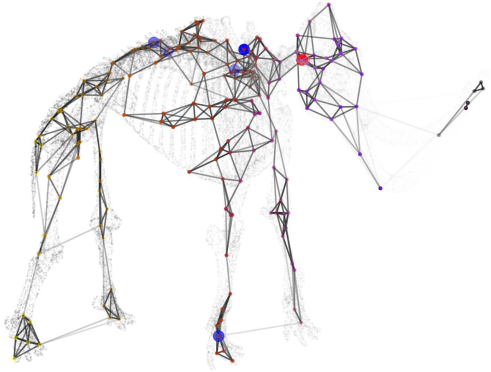
Optimization with **Stochastic Gradient Descent**

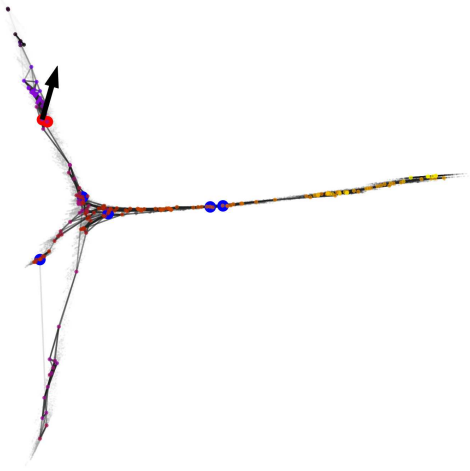
Fitting low dimensional representation





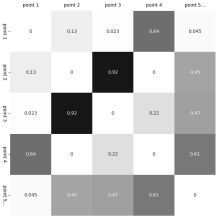
Fitting low dimensional representation

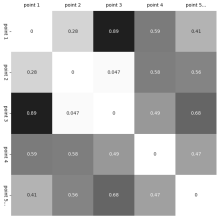


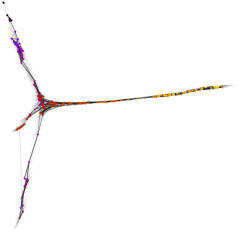


Fitting low dimensional representation

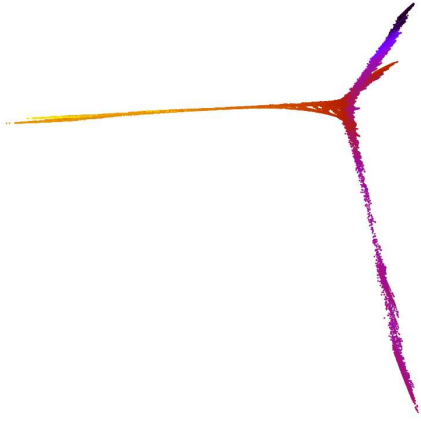




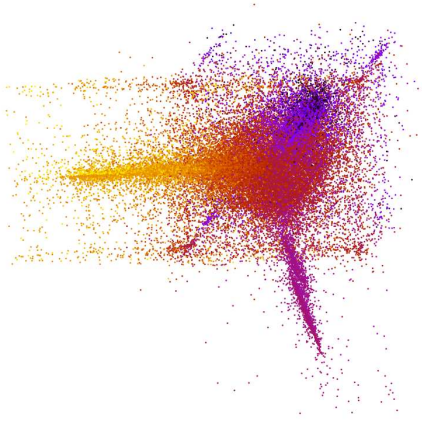




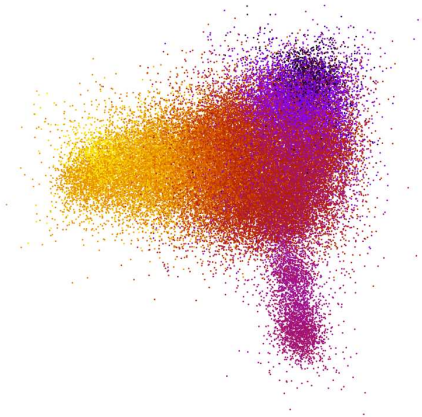
Optimization



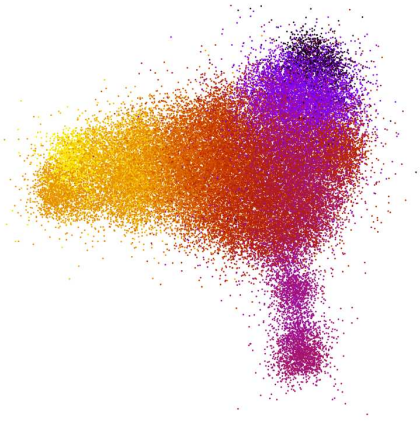
Optimization



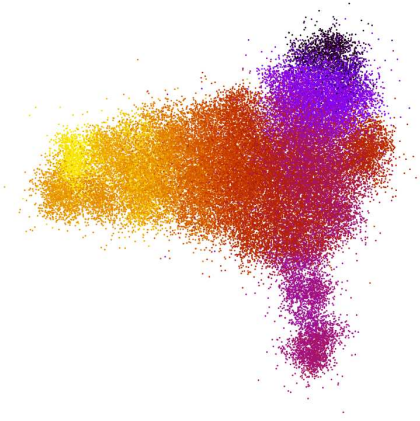
Optimization



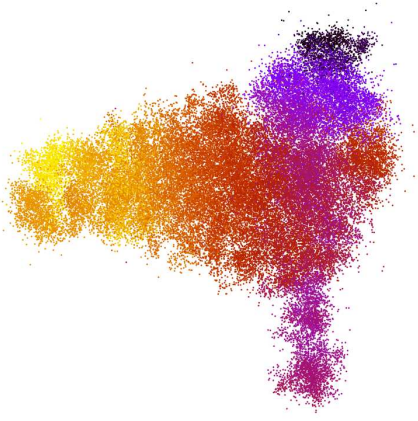
Optimization



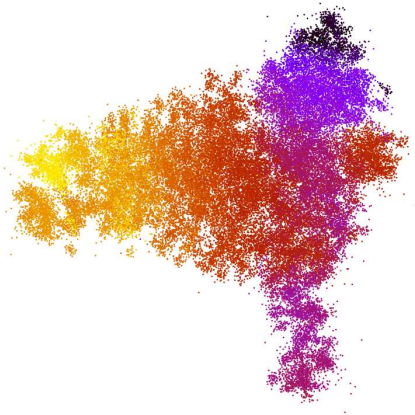
Optimization



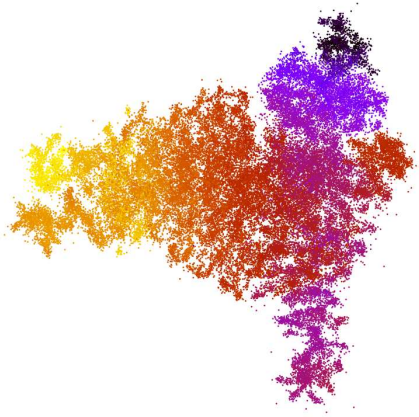
Optimization



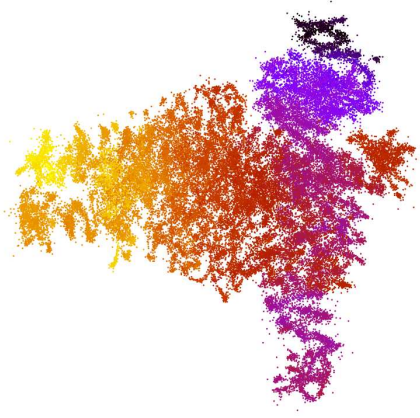
Optimization



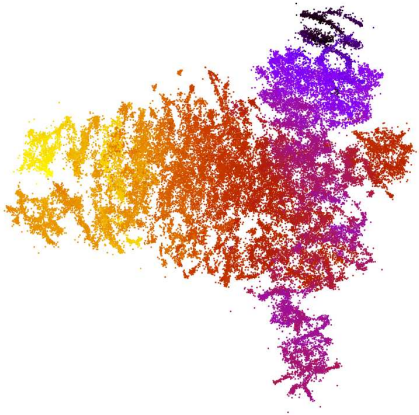
Optimization



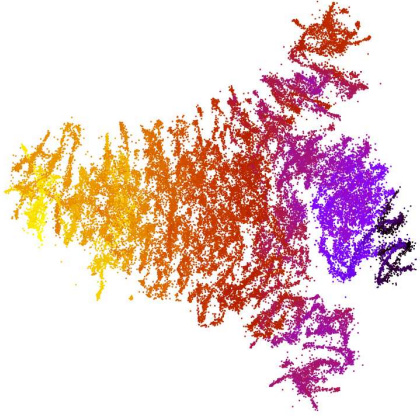
Optimization



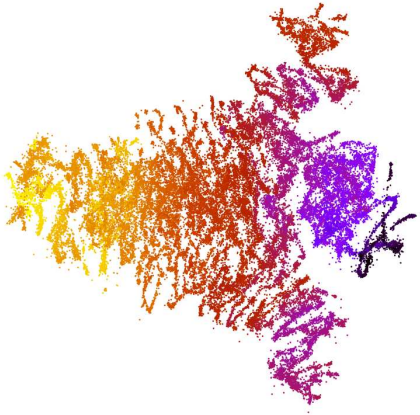
Optimization



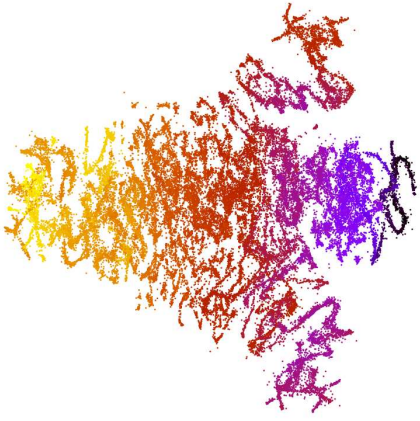
Optimization



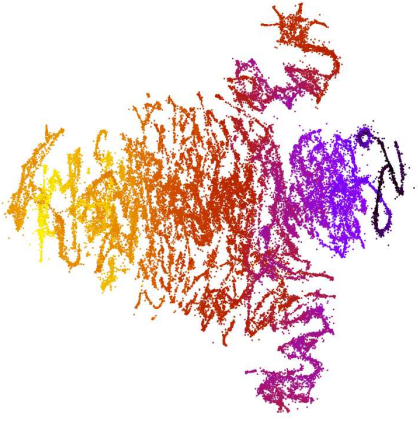
Optimization



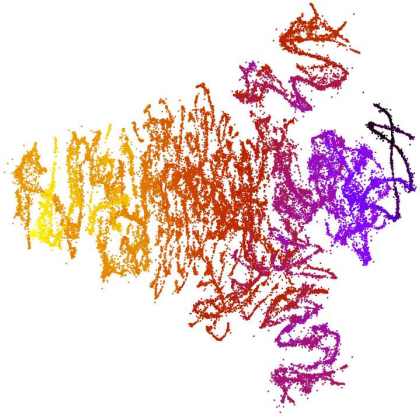
Optimization



Optimization



Optimization



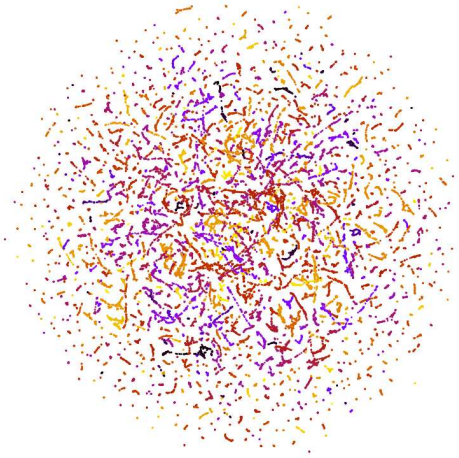
Optimization



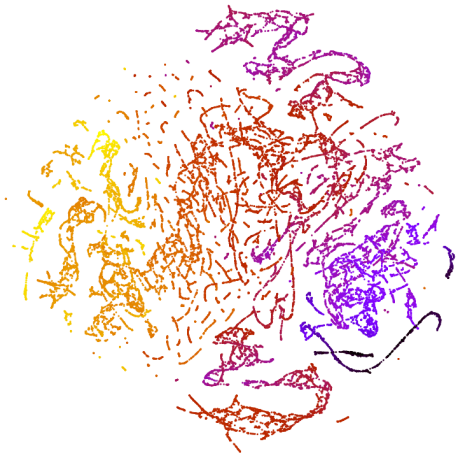
Basic parameters

Parameter	Meaning	Default value
`n_components`	target dimensions	`2`
`n_neighbors`	influence of neighborhood	`15`
`min_dist`	how points are packed	`0.1`
`metric`	how distance is measured	`euclidian`

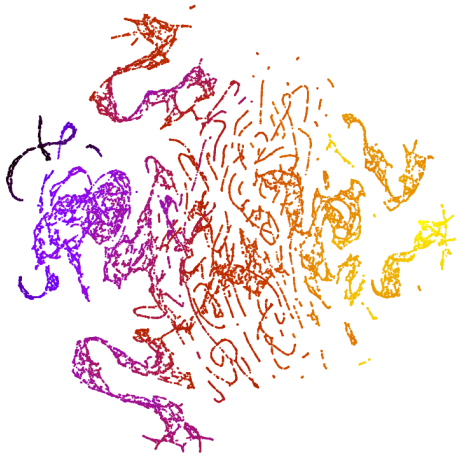
Influence of the number of neighbors



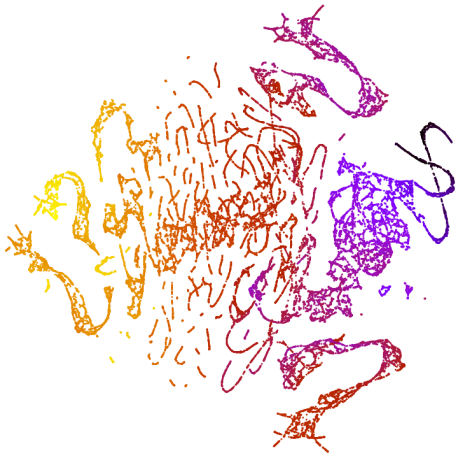
Influence of the number of neighbors



Influence of the number of neighbors



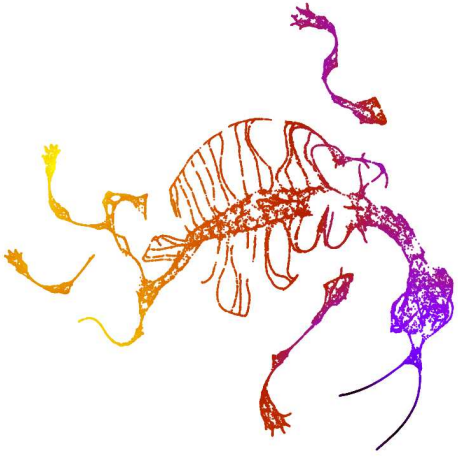
Influence of the number of neighbors



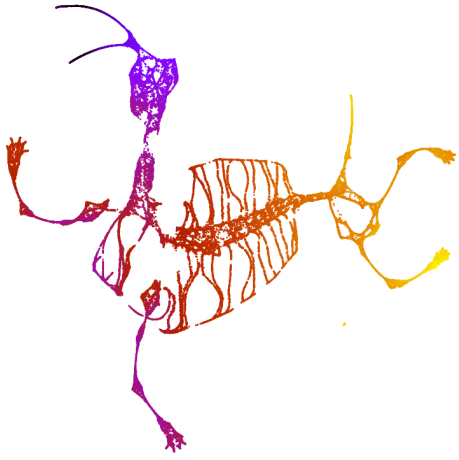
Influence of the number of neighbors



Influence of the number of neighbors



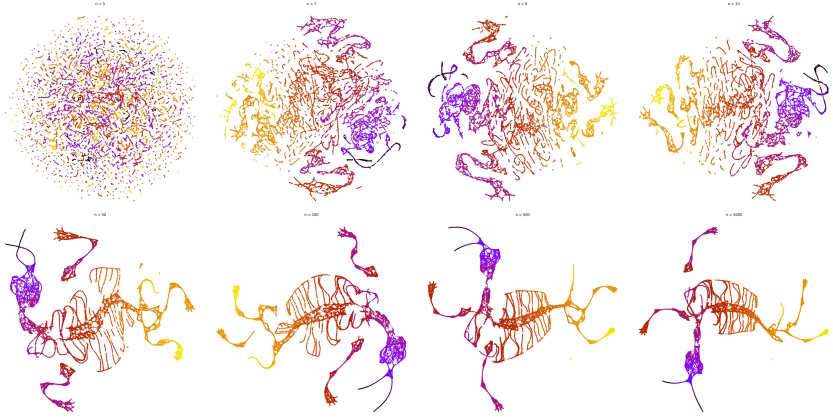
Influence of the number of neighbors



Influence of the number of neighbors



Influence of the number of neighbors



UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic

T-SNE

- probabilities

UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic
- determinist initialization

T-SNE

- probabilities
- random initialization

UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic
- determinist initialization
- log_2 similarity scores

T-SNE

- probabilities
- random initialization
- gaussian distribution similarity

UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic
- determinist initialization
- log_2 similarity scores
- update by pairs

$\rightarrow$ scales well for large datasets

T-SNE

- probabilities
- random initialization
- gaussian distribution similarity
- update all points

scales less

Scaling

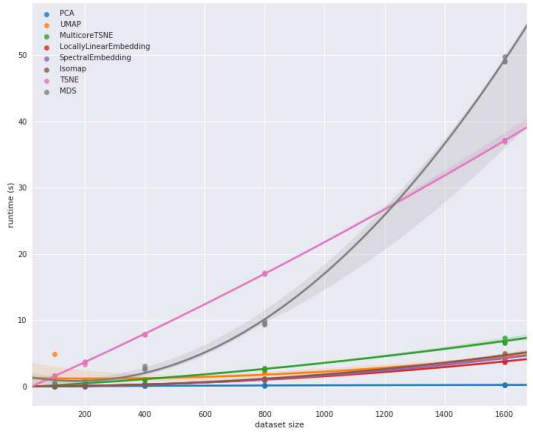


Figure from umap-learn documentation

Possible traps

- Hyperparameters matter (see `n_neighbors` examples)

Possible traps

- Hyperparameters matter (see `n_neighbors` examples)
- Absolute distances mean nothing (relative do in some way)

Possible traps

- Hyperparameters matter (see `n_neighbors` examples)
- Absolute distances mean nothing (relative do in some way)
- Cluster sizes mean nothing (uniformization)

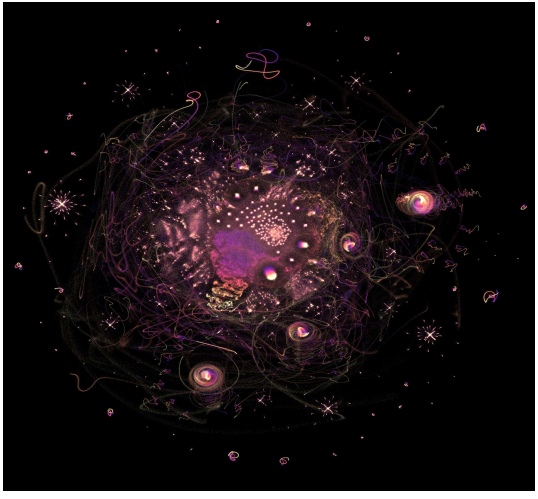
Possible traps

- Hyperparameters matter (see `n_neighbors` examples)
- Absolute distances mean nothing (relative do in some way)
- Cluster sizes mean nothing (uniformization)
- In high dimension, noise can have shape (carefull with `n_neighbors` values)

Possible traps

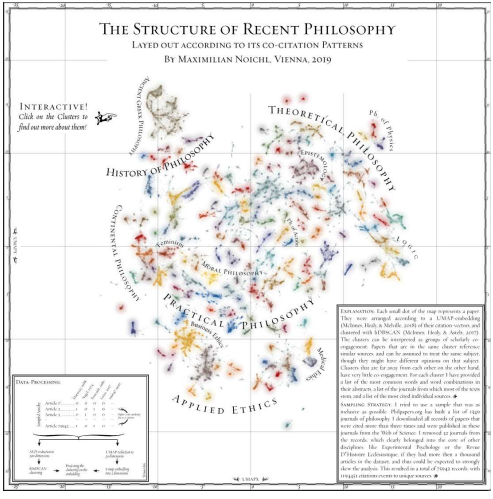
- Hyperparameters matter (see `n_neighbors` examples)
- Absolute distances mean nothing (relative do in some way)
- Cluster sizes mean nothing (uniformization)
- In high dimension, noise can have shape (carefull with `n_neighbors` values)
- Algorithm is stochastic

Cool examples



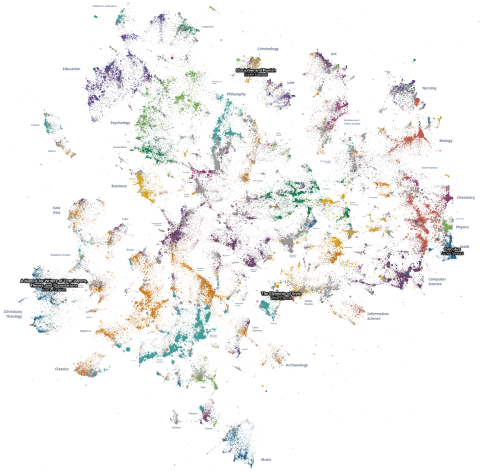
1 million integers by divisibility

Cool examples



Structure of Recent Philosophy

Cool examples



Open Syllabus Galaxy

Cool examples

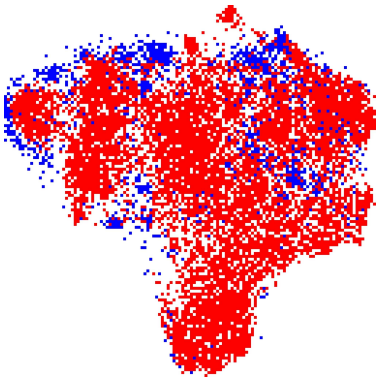






H. Goëau - TDWG 2021

Cool examples

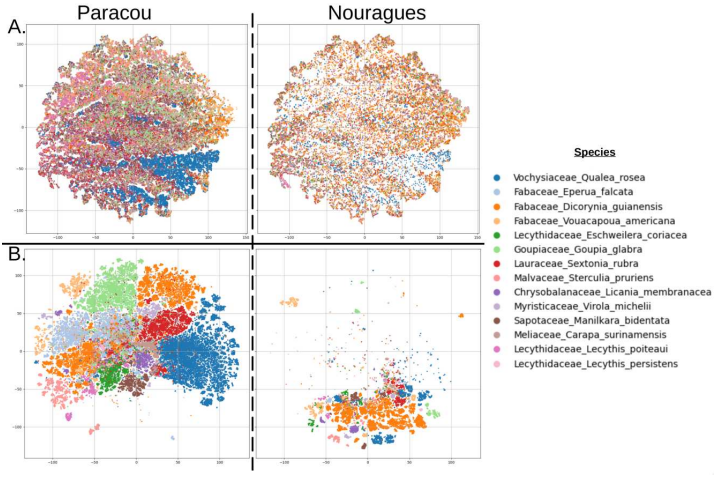






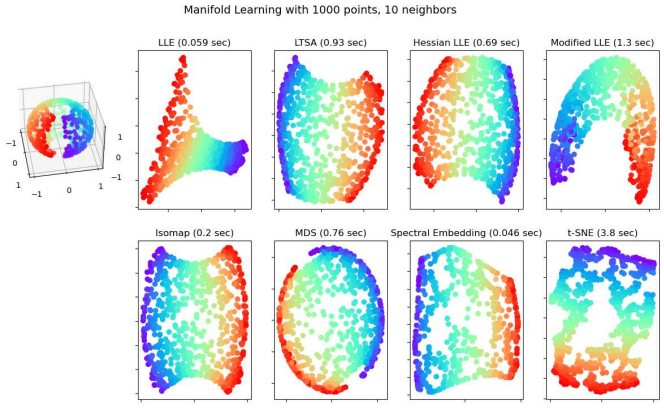
H. Goëau - TDWG 2021

Cool examples



C. Prieur PhD thesis

A lot of options !



Useful ressources

- `umap-learn`
- `scikit-learn`
- Topos institute - understanding UMAP
- Google PAIR - understanding UMAP

on YouTube:

- Talk by Leland McInnes
- deepia
- StatQuest
- DeepFindr

Thanks for your attention !

Let’s practice !

References i

Du, Trina Y (2019). **“Dimensionality reduction techniques for visualizing morphometric data: Comparing principal component analysis to nonlinear methods”.** In: *Evolutionary Biology* 46.1, pp. 106–121.

Hinton, Geoffrey E and Sam Roweis (2002). **“Stochastic neighbor embedding”.** In: *Advances in neural information processing systems* 15.

McInnes, Leland, John Healy, and James Melville (2018). **“Umap: Uniform manifold approximation and projection for dimension reduction”.** In: *arXiv preprint arXiv:1802.03426*.

Van der Maaten, Laurens and Geoffrey Hinton (2008). **“Visualizing data using t-SNE.”.** In: *Journal of machine learning research* 9.11.